MARKETING ANALYTICS REGRESSION REPORT

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Explaining Customer Purchase Amount

1. Key Findings

Why Model 2 is the strongest model

The clean sample contains 4,861 customers after the fixed 1.5 x IQR rule removes 139 unique records (2.78%). Model 2 explains PurchaseAmount with Income, PromoUsed, PagesViewed, and WebsiteVisits. Exploratory plots show clear positive relationships with purchasing, while correlations among these predictors are negligible. CustomerID is excluded as an identifier, and Churn is excluded because its timing relative to purchase is unknown.

Model 2 clearly improves on the Income-only baseline. Model 1 explains 32.33% of purchase variation with RMSE = 17.26; Model 2 explains 76.99% (adjusted R-squared = .7697) and reduces RMSE to 10.06, a 41.7% improvement. It is jointly significant, $F(4, 4856) = 4062.85$, $p < .001$, with substantially lower AIC and BIC. Model 3's promotion-by-pages interaction is not significant ($p = .171$), adds virtually no R-squared, and worsens AIC and BIC. Model 2 therefore offers the strongest combination of accuracy and clarity.

$$PurchaseAmount = 0.7728 + 0.000496(Income) + 20.0111(PromoUsed) + 1.4700(PagesViewed) + 1.9934(WebsiteVisits)$$

The equation estimates expected purchase amount while holding the other predictors constant. RMSE = 10.06 means individual predictions commonly differ from actual purchases by about 10 dataset units, making the model more suitable for planning and group-level forecasting than exact individual promises. The intercept is not significant ($p = .323$) and has no useful marketing interpretation.

2. Interpreting the Strongest Model

Which variables matter most, and why?

Income matters most statistically. Its standardized coefficient is .587, and removing it reduces R-squared by .344. Holding the other variables constant, 10,000 additional income units are associated with 4.96 purchase units (95% CI [4.85, 5.08]). Customers with greater purchasing capacity tend to spend more, making Income useful for broad segmentation, not a causal or controllable marketing lever.

Promotion use ranks second (standardized beta = .477; unique R-squared contribution = .227). Promotion users purchase 20.01 more units than otherwise similar non-users (95% CI [19.44, 20.58]). It is the largest controllable group difference, but it may partly reflect who receives or chooses promotions. The result supports an experiment, not a claim of guaranteed promotion lift.

Each additional page viewed is associated with 1.47 purchase units, and each visit with 1.99 units. Their standardized betas are .383 and .291. Two additional visits plus three additional pages correspond to about 8.40 purchase units, holding Income and PromoUsed constant. This is consistent with deeper exploration and purchase intent, although intended spending may also cause more browsing.

All four slopes have $p < .001$ and narrow confidence intervals. Income explains the most variation, PromoUsed provides the clearest actionable signal, and engagement identifies behaviour that the customer journey may influence. The promotion-by-pages interaction is not significant, so separate page-view effects are unsupported. The median residual is 0.01, indicating no systematic overprediction or underprediction.

3. Marketing Implications

Recommendation 1 - Test promotions before scaling them

Promotion users purchase 20.01 more units after Income and engagement are held constant. Because PromoUsed is observational, the decision is not to discount for everyone; it is to test whether the offer creates incremental, profitable demand. Randomly assign otherwise eligible customers to the promotion or current offer, keeping product mix, campaign period, and channel comparable.

Evaluate incremental purchase amount, conversion, gross margin, redemption, and incentive cost. Expand only when incremental contribution margin exceeds discount, media, and fulfilment costs. Report broad income and engagement bands while preserving random assignment within each band. This converts the 20.01-unit association into a measurable decision without treating it as causal lift.

Recommendation 2 - Improve the digital exploration journey

One additional page is associated with 1.47 purchase units and one visit with 1.99 units; two more visits plus three more pages correspond to about 8.40 units. Prioritize changes that help customers find relevant products and continue their journey: faster landing pages, clearer categories, better search and filters, useful recommendations, saved carts, and consistent cross-device experiences.

Test these changes and retain only those that increase purchase amount or profit, not merely page views. Track conversion, purchase amount, bounce rate, search success, cart completion, and repeat visits. Browsing may be both a driver and a result of purchase intention; more clicks without more profitable purchasing are not success.

Recommendation 3 - Use income for broad, ethical segmentation

A 10,000-unit income difference is associated with 4.96 purchase units. Use consented income bands to tailor broad value propositions: test affordability, bundles, or payment flexibility in lower-income segments and premium assortment or convenience in higher-income segments. These are hypotheses for message and assortment tests, not assumptions about individuals.

Do not use Income for discriminatory pricing, automatic exclusion, or judgments about one customer's willingness to pay. Compare reach, offer availability, acceptance, and outcomes across income and relevant protected groups. Income is not a marketing lever, and the model does not show that it causes spending; use it only for responsible, broad planning.

Recommendation 4 - Use Model 2 for planning and prioritization

Model 2 explains 76.99% of purchase variation, has RMSE = 10.06, and remains stable when all 139 removed observations are restored: no slope changes by more than 1.32%. Use it to rank broad groups or campaign scenarios, forecast aggregate purchase amount under different promotion and engagement assumptions, and plan inventory, service capacity, and budgets. Group forecasts are more dependable than exact individual predictions.

Report prediction intervals and scenario ranges rather than one exact number. Validate on later customers and monitor RMSE, bias, segment-level error, and profit. The non-significant promotion-by-pages interaction keeps implementation simple: one additive benchmark is sufficient until experiments or newer data show a meaningful segment-specific effect.

5. Limitations

Data quality and sample treatment

There are no missing cells, duplicate rows, or duplicate CustomerID values, and cleaned PurchaseAmount is nearly symmetric (skewness = 0.04). The main quality concern is outlier treatment: 139 customers (2.78%), including 25 PurchaseAmount outliers, are removed. The restored-outlier slopes change by at most 1.32%, supporting robustness, but rare high-value or high-activity behaviour may still be underrepresented. Thirty-two Clicks-greater-than-AdImpressions records also require source review; those fields were excluded rather than guessed at.

Functional form is the main assumption weakness. Although the residual plots are mostly straight and individual squared terms do not improve fit, RESET is significant, $F(2, 4854) = 3.41$, $p = .033$. A combined curve, omitted interaction, or omitted variable may make constant slopes slightly misleading; GAMs, restricted splines, and justified interactions should be tested. Other results are satisfactory: Shapiro-Wilk $p = .692$, Breusch-Pagan $p = .581$, VIF values are near 1, and maximum Cook's D is .00415. Durbin-Watson = 2.033 shows no row-order pattern but cannot prove independent sampling.

Model 2 is cross-sectional and observational. Promotion use may reflect prior interest, and greater purchase intention may cause more browsing, so coefficients are not causal. Product, price, discount depth, channel, exposure, prior purchases, tenure, seasonality, margin, and competitors are omitted; 23.01% of variation remains unexplained, and no holdout or later-period validation was performed. Currency, observation window, and the exact meaning of PurchaseAmount are undefined. Income is sensitive and non-actionable, and individual forecasts are uncertain. Use explicit units, prediction intervals, fairness review, and a clearly defined target population.

6. Future Improvements

Additional variables needed

Define the purchase window and currency, then add recency, frequency, prior spending, basket and product details, price, discount depth, promotion exposure and redemption, channel, device, tenure, loyalty, margin, fulfilment cost, returns, inventory, seasonality, region, and market conditions. These variables can explain omitted demand and support profit rather than purchase amount alone. Collect sensitive fields only with purpose, consent, access controls, and fairness monitoring.

Alternative models and segmentation

Test a generalized additive model first because RESET suggests missing functional-form detail; smooth terms can capture curves while remaining interpretable. Use ridge, lasso, or elastic net when more correlated predictors are added. Random forests and boosted trees can discover thresholds and interactions, but require unseen-data testing and partial-dependence or SHAP explanations.

Use clustering on recency, frequency, monetary value, and engagement to create stable, sizeable, reachable segments, then compare segment-specific regressions or interactions. Clustering supports segmentation rather than replacing prediction. With randomized promotion data, use uplift or causal models to identify customers who benefit incrementally from an offer.

Improving and validating predictive accuracy

Use a later-period test set, or reserve an untouched test sample and tune only through training-set cross-validation. Compare RMSE, MAE, R-squared, interval coverage, segment bias, and profit; check leakage, transformations, interactions, and outlier sensitivity. Monitor data quality, error, fairness, and drift after deployment. Keep Model 2 as the transparent benchmark and replace it only for a repeatable unseen-customer improvement that supports better decisions.

Appendix A. Statistical Tables

Tables and figures are placed in appendices because the assignment excludes them from the six-page narrative limit. Values are reproduced from the submitted analysis workflow.

Table A1. Data Quality and IQR Outlier Profile

Quality check	Count	Analytical treatment
Original rows	5,000	Records before outlier removal
Rows removed	139	2.78% flagged by 1.5 x IQR
Analysis rows	4,861	Used in all primary models
Columns	15	Fields in supplied workbook
Missing cells	0	No imputation required
Duplicate rows	0	No deletion required
Duplicate CustomerID	0	Identifier is unique
Clicks > AdImpressions	32	Flagged; two removed by IQR, fields not modelled

Note. The outlier union removes a row once even when it is flagged on several continuous fields.

Table A1 (continued). IQR Outlier Screen

Variable	Lower fence	Upper fence	Count	Percent
Age	-10.00	102.00	0	0.00%
Income	21177.27	158073.01	17	0.34%
WebsiteVisits	2.00	18.00	37	0.74%
TimeOnSite	-0.55	10.44	12	0.24%

Variable	Lower fence	Upper fence	Count	Percent
PagesViewed	-10.00	30.00	0	0.00%
AdImpressions	-254.00	754.00	0	0.00%
Clicks	-1.00	7.00	53	1.06%
EmailOpens	-15.50	44.50	0	0.00%
SatisfactionScore	-0.97	7.00	0	0.00%
PurchaseAmount	31.73	147.75	25	0.50%

Note. Binary and categorical variables are excluded from the IQR rule. The union of flagged rows was removed before EDA, model fitting, and primary diagnostics.

Table A2. Selected-Variable Pearson Correlation Matrix

Variable	Purchase	Income	Promo	Pages	Visits
PurchaseAmount	1.0000	0.5686	0.4423	0.3689	0.2876
Income	0.5686	1.0000	-0.0410	-0.0035	0.0072
PromoUsed	0.4423	-0.0410	1.0000	-0.0199	-0.0097
PagesViewed	0.3689	-0.0035	-0.0199	1.0000	-0.0086
WebsiteVisits	0.2876	0.0072	-0.0097	-0.0086	1.0000

Note. No selected predictor pair approaches the |.70| screening threshold. VIF results appear in Table A8.

Table A3. Model Fit Comparison

Model	R2	Adj. R2	RMSE	RSE	AIC	BIC	F	df	p
Model 1	0.3233	0.3232	17.26	17.26	41493.19	41512.66	2321.54	1, 4859	<2.2e-16
Model 2	0.7699	0.7697	10.06	10.07	36254.82	36293.75	4062.85	4, 4856	<2.2e-16
Model 3	0.7700	0.7698	10.06	10.07	36254.95	36300.37	3251.24	5, 4855	<2.2e-16

Note. RMSE uses the mean squared residual denominator n; RSE uses residual degrees of freedom. Lower AIC, BIC, and RMSE are preferred.

Table A4. Model 1 Coefficients and Confidence Intervals

Term	Estimate	SE	t	p	95% CI low	95% CI high
Intercept	46.6087	0.9260	50.332	<2.2e-16	44.7932	48.4241
Income	0.000480	0.000010	48.182	<2.2e-16	0.000461	0.000500

Note. Outcome: PurchaseAmount. Confidence intervals are two-sided at 95%.

Table A5. Model 2 Coefficients and Confidence Intervals

Term	Estimate	SE	t	p	95% CI low	95% CI high
Intercept	0.7728	0.7818	0.988	0.323	-0.7600	2.3056
Income	0.000496	0.000006	85.267	<2.2e-16	0.000485	0.000508
PromoUsed	20.0111	0.2892	69.204	<2.2e-16	19.4442	20.5780
PagesViewed	1.4700	0.0264	55.619	<2.2e-16	1.4182	1.5218
WebsiteVisits	1.9934	0.0471	42.310	<2.2e-16	1.9011	2.0858

Note. Outcome: PurchaseAmount. Confidence intervals are two-sided at 95%.

Table A6. Model 3 Coefficients and Confidence Intervals

Term	Estimate	SE	t	p	95% CI low	95% CI high
Intercept	15.3432	0.7334	20.920	<2.2e-16	13.9053	16.7810
Income	0.000496	0.000006	85.264	<2.2e-16	0.000485	0.000508
PromoUsed	20.0111	0.2891	69.210	<2.2e-16	19.4442	20.5779
PagesViewed_c	1.4342	0.0372	38.554	<2.2e-16	1.3612	1.5071
WebsiteVisits	1.9930	0.0471	42.303	<2.2e-16	1.9006	2.0853
Promo x Pages_c	0.0723	0.0529	1.368	0.171	-0.0313	0.1759

Note. Outcome: PurchaseAmount. Confidence intervals are two-sided at 95%.

Table A7. Model 2 Standardized Effects and Unique Contribution

Predictor	Standardized beta	Unique delta R2	Partial R2
Income	0.587	0.344	0.600
PromoUsed	0.477	0.227	0.497
PagesViewed	0.383	0.147	0.389
WebsiteVisits	0.291	0.085	0.269

Note. Unique delta R2 is the decrease in full-model R2 after removing one predictor.

Standardized betas rank effects measured on different scales.

Table A8. Required Diagnostic Tests

Model	W	Shapiro p	BP	BP p	RESET F	RESET p	Max Cook D	>4/n
Model 1	0.9977	1.38e-06	0.813	0.367	2.043	0.130	0.00308	228
Model 2	0.9997	0.692	2.860	0.581	3.406	0.033	0.00415	237
Model 3	0.9997	0.665	2.738	0.740	3.093	0.045	0.00359	245

Note. For Model 2: Shapiro-Wilk p = .692, Breusch-Pagan p = .581, and RESET p = .033. RESET is reported as a mild functional-form warning; the 4/n Cook rule is a review threshold.

Table A8 (continued). Model 2 Variance Inflation Factors

Predictor	VIF	Tolerance
Income	1.0017	0.9983
PromoUsed	1.0022	0.9978
PagesViewed	1.0005	0.9995
WebsiteVisits	1.0002	0.9998

Note. VIF values near 1 indicate negligible coefficient inflation from joint predictor overlap.

Table A8 (continued). Targeted Quadratic Checks

Added term	Estimate	p	Change in R2	Change in AIC
Income squared	0.000000	0.531	0.000019	1.61

Added term	Estimate	p	Change in R2	Change in AIC
PagesViewed squared	0.003615	0.504	0.000021	1.55
WebsiteVisits squared	0.001591	0.893	0.000001	1.98

Note. None of the squared terms is significant, and every positive AIC change indicates a worse complexity-adjusted fit than Model 2.

Table A9. Outlier Sensitivity Refit

Term	Cleaned sample	Restore all outliers	Slope change
Intercept	0.7728	-0.0139	-
Income	0.000496	0.000503	1.32%
PromoUsed	20.0111	20.0105	-0.00%
PagesViewed	1.4700	1.4802	0.70%
WebsiteVisits	1.9934	2.0017	0.42%

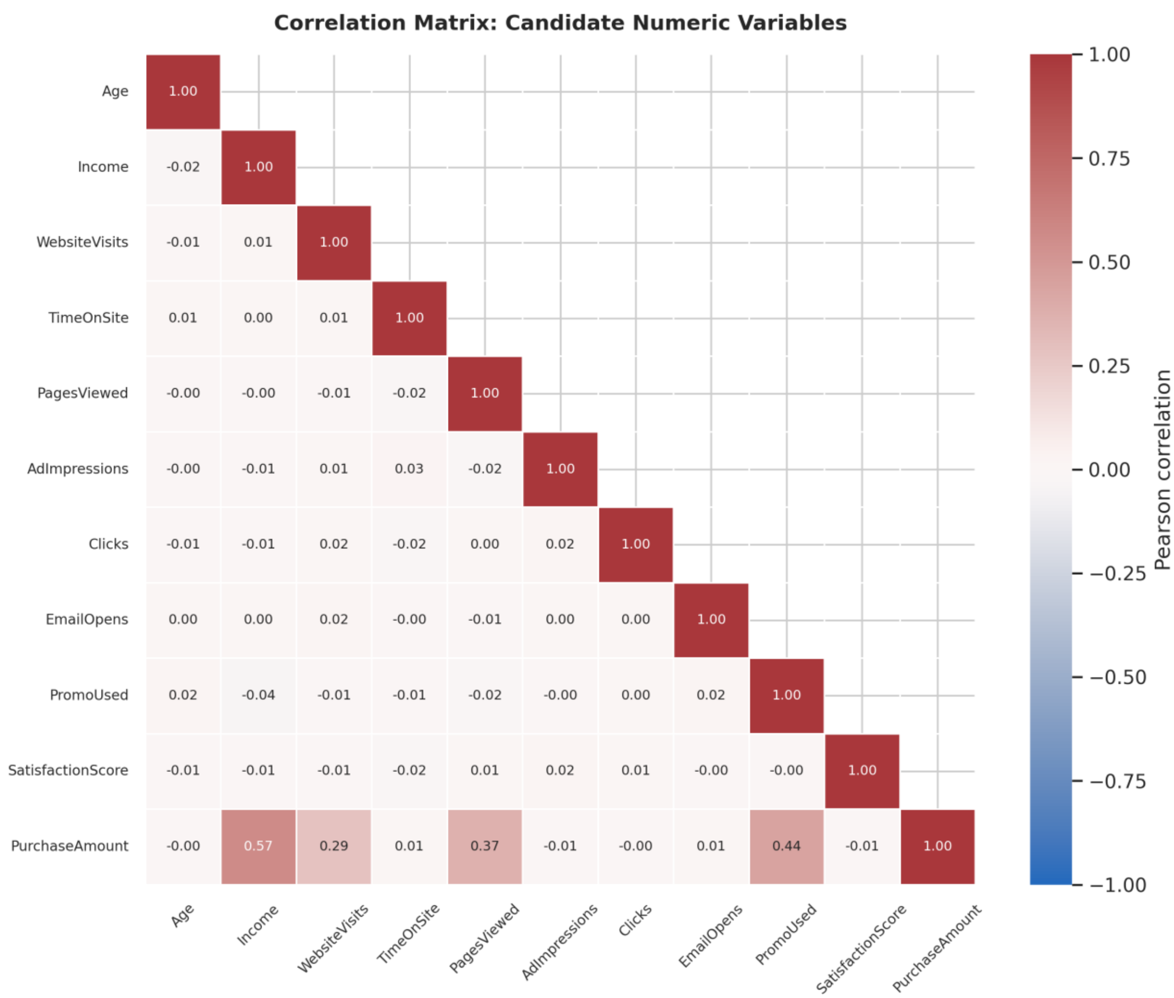
Note. The intercept percentage is omitted because it is not a meaningful business slope.

Restoring the 139 rows changes every slope by at most 1.32%; R2 changes from 0.7699 to 0.7769, and RMSE from 10.06 to 10.13.

Appendix B. Figures

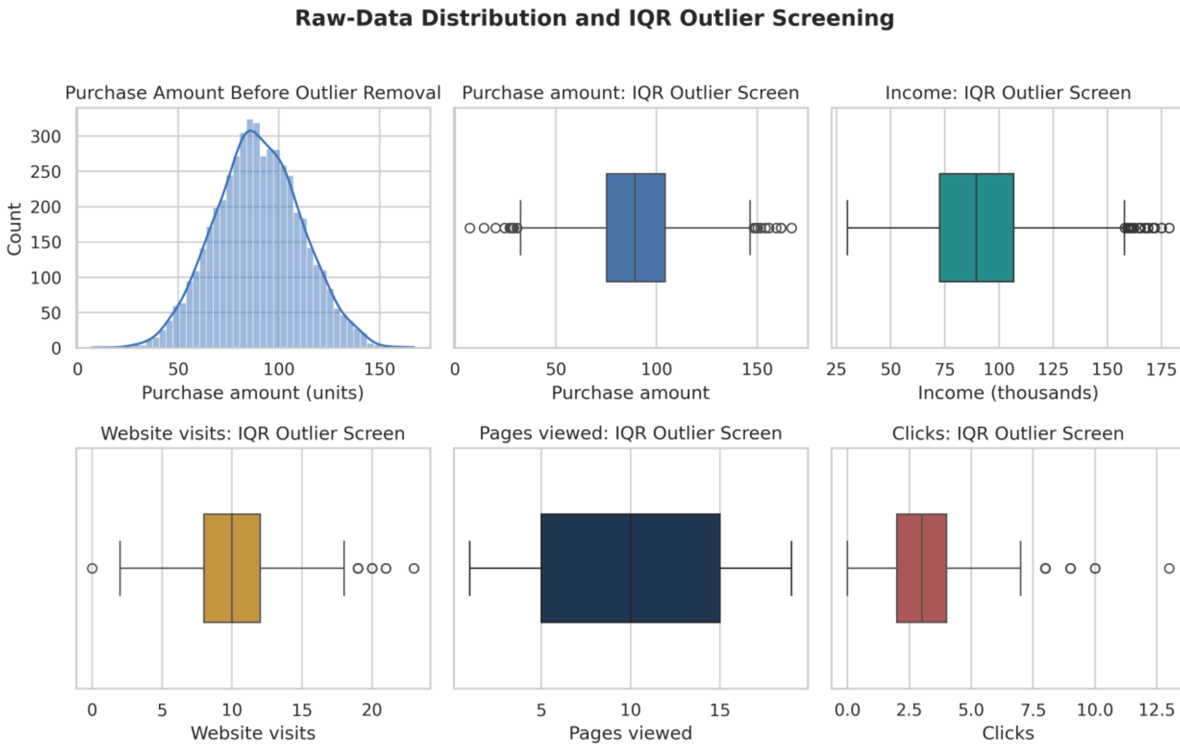
The following figures document the exploratory evidence and preferred-model diagnostics discussed in the six-page narrative.

Figure B1. Candidate-variable correlation matrix



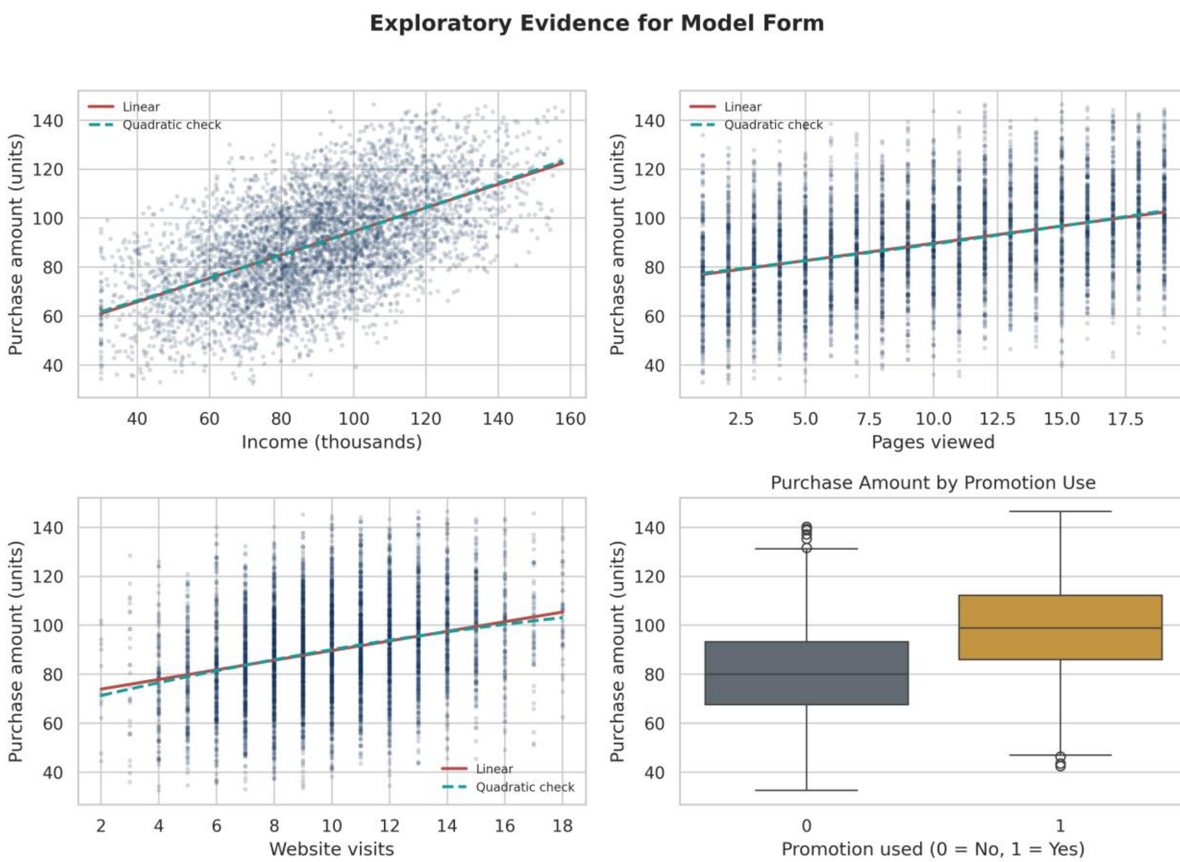
Note. Pearson correlations. PurchaseAmount is most strongly related to Income, PromoUsed, PagesViewed, and WebsiteVisits; candidate predictors are otherwise nearly uncorrelated.

Figure B2. Distribution and outlier screening



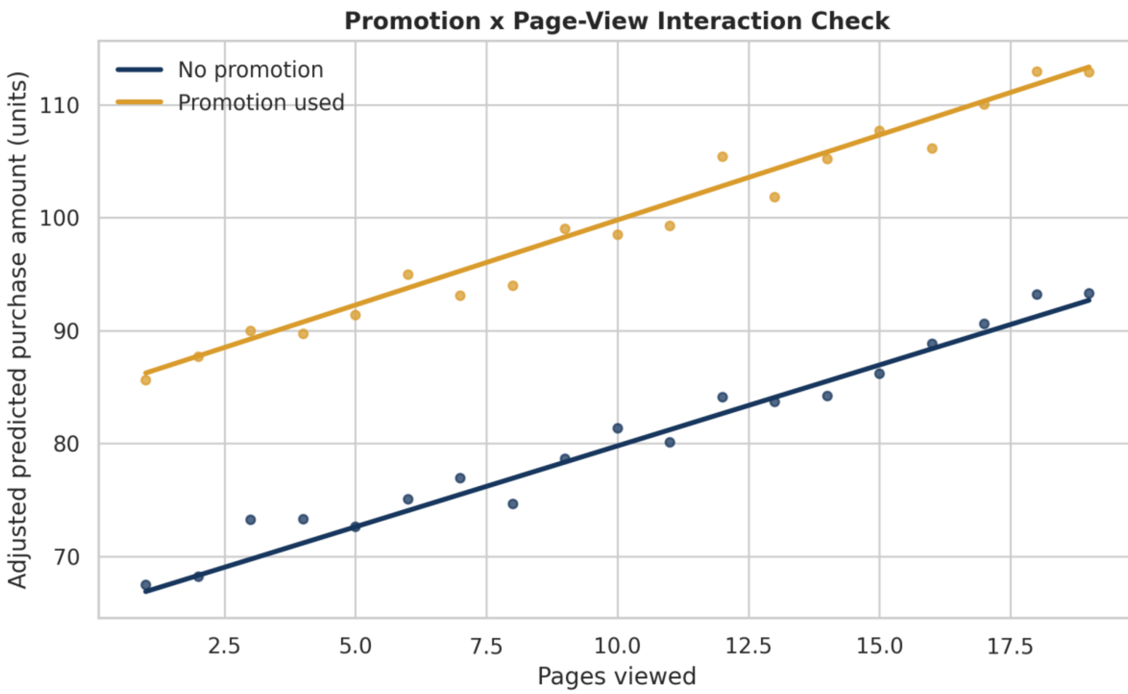
Note. The original-sample plots show the sparse extreme values that produced the 139-row union removed before modelling.

Figure B3. Exploratory evidence for model form



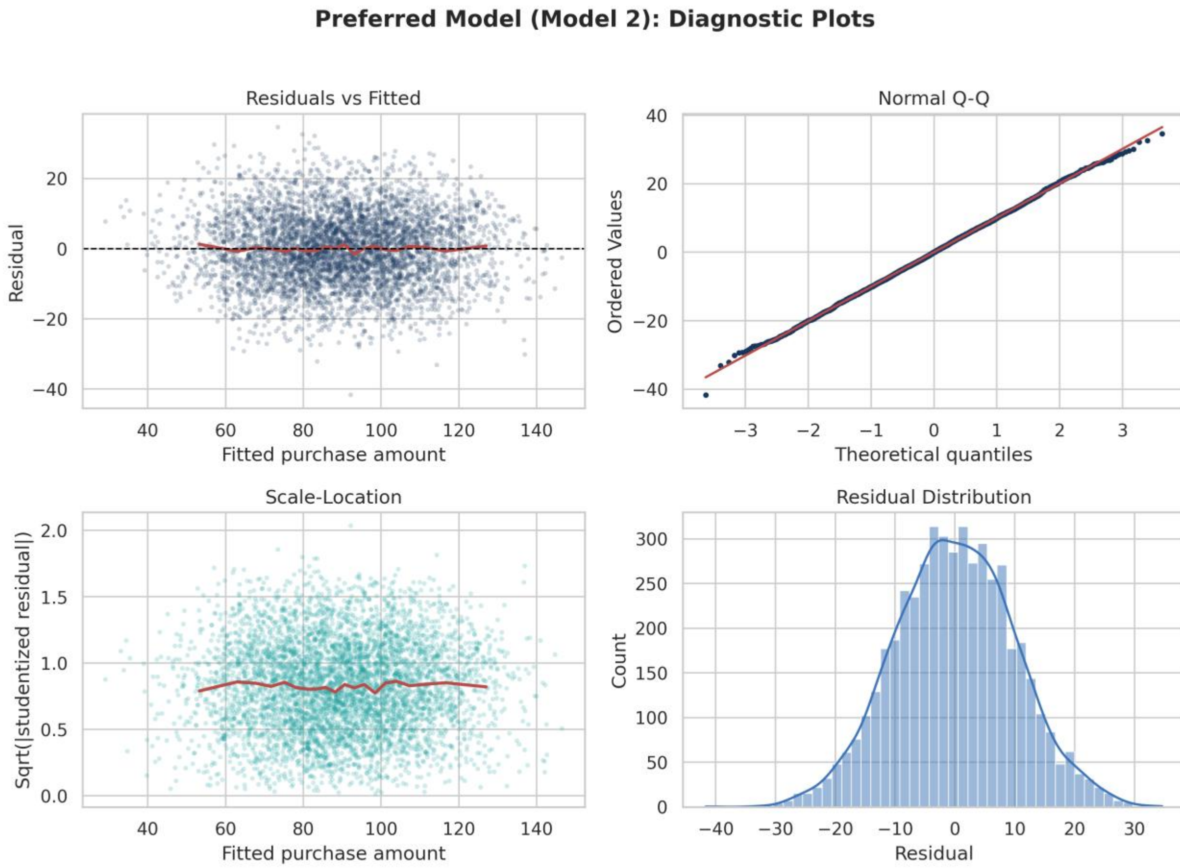
Note. Linear and quadratic checks are nearly coincident for the quantitative predictors; promotion use shifts the purchase distribution upward.

Figure B4. Promotion by page-view interaction check



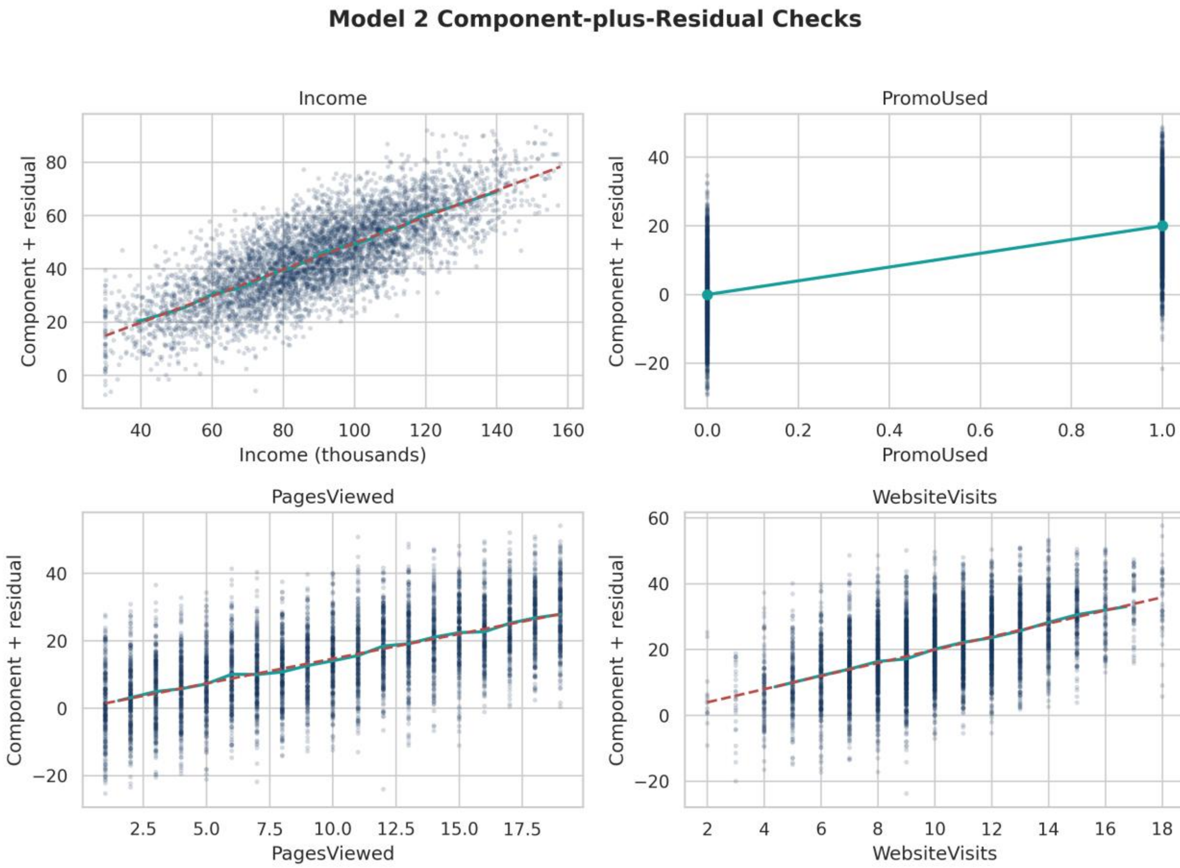
Note. The adjusted fitted lines are nearly parallel. Model 3 confirms that their slope difference is not significant (p = .171).

Figure B5. Preferred-model diagnostic plots



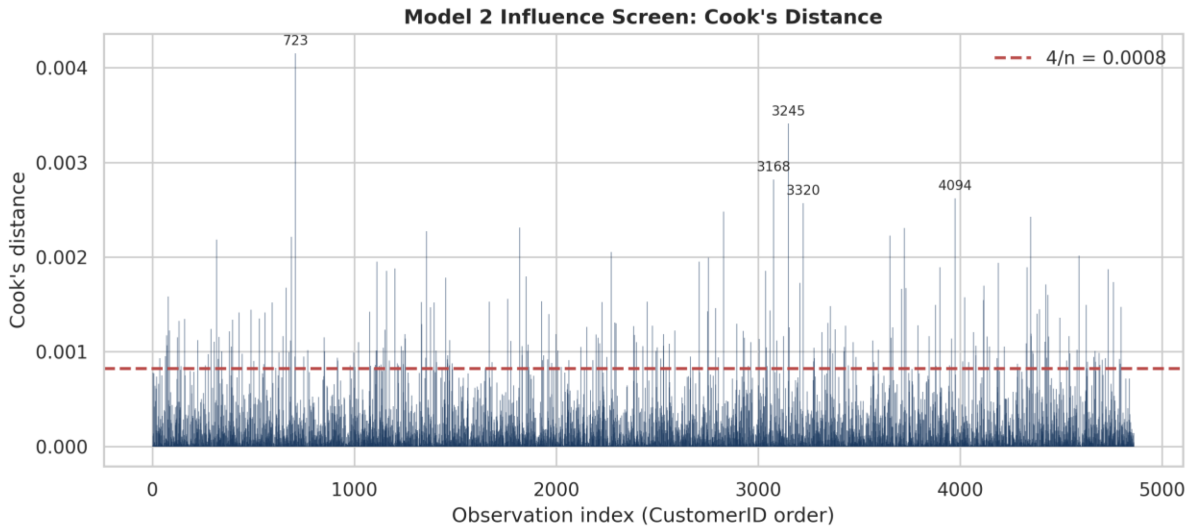
Note. Residuals are centred without a strong fan-shaped spread, and the Q-Q plot and density support approximate normality; RESET still gives a mild functional-form warning.

Figure B6. Model 2 component-plus-residual checks



Note. The adjusted smoothers show only small departures from their linear components; targeted squared terms do not improve the model.

Figure B7. Cook's distance influence screen



Note. The sensitive 4/n rule flags review candidates, but maximum Cook's D is .00415 and no remaining observation is materially dominant.